Group coursework worth 100% of your final mark
Software Engineering and Professional Practice and Building Usable Software
School of Computer Science, University of Birmingham, UK and Dubai
(In collaboration with the University Estates, Siemens and Exchange)

Submission is December 6th 2021, 11:59 p.m. UK via Canvas (Firm Deadline)

I. Challenge Description: Pick ONLY ONE of the below three Challenges.

Challenge 1 brief description – "Living Lab":

The University of Birmingham, in partnership with Siemens, is combining digital sensor and analytics technologies, artificial intelligence, decentralised energy generation and storage, renewable energy and concepts that help change users' behaviours transform the University's Edgbaston and Dubai campuses into the world's smartest global campus, creating a 'Living Lab'. Starting in Autumn 2021, the first phase of the Living Lab includes a major energy efficiency project to roll out of 23,000 Enlighted IoT sensors across the University estate and to become the first university in the world to roll out Internet of Things (IoT) technology at scale.

Figure 1. Examples of Enlighted IoT Sensors

The 'Living Lab' will capture data from the University's buildings, technologies used on campus, estates infrastructure and energy plants and use the data for innovation, R&D activities, energy monitoring for as well as teaching.

For Software Engineers, sensors can capture live data about energy demands, production, and savings; motion; temperature; estates/offices/labs usage and utilisation etc. to provide a unique opportunity for innovating data-driven software architecture

and applications of societal and climate benefits. These applications can be of functional use by the university community, management, research and public. Applications may possibly have futuristic and blue skies characters. They can also support research (i.e. Research Software Engineers). Applications may capture, monitor, analyse, visualise, simulate data. Applications may use gamification, machine learning, data analysis and dynamic optimisation techniques to change users' behaviours on campus; better optimise operations on campus to reach Governmental NetZero emissions targets; and to provide opportunities for raising public awareness etc.

We are seeking innovative suggestions for applications of your choice: these could be of direct use or of futuristic nature. *The scope is open and can relate to any application you can envision.*

Examples of applications for Challenge 1 may include, but not limited to:

- Buildings tweets: building tweeting in real time about its energy consumption; contributions towards NetZero targets; flow of users, utilization etc.
- Apps for heatmapping and visualizing energy consumption in buildings and/or across campus;
- Apps for university planning to optimize space utilization;
- Apps that use that use machine learning for predicting usage and to inform what-if analysis;
- Apps that use gamification, informed by the data, to change behavior – e.g., provide incentives for better utilization of space and to reduce energy consumption etc;
- Apps for governmental reporting of energy usages;
- Digital twins and simulation systems for live monitoring, tracking and visualization of campus facilities and what-if analysis etc.
- Fun applications that makes use of the infrastructure and data, such as:
    - daily/weekly/monthly competition between buildings for sustainability; applications for casting energy consumption and savings into music tunes on campus; applications for casting energy consumption and savings into light/patterns to be projected on campus buildings, old Joe etc; coordinated drones above campus buildings celebrating reduction of energy usage etc.

Figure 2: Example of existing applications

More References that relate to Challenge 1 brief description:
Ref1.https://www.birmingham.ac.uk/university/building/smart-campus/index.aspx
Ref2.https://www.birmingham.ac.uk/documents/university/smart-campus/the-smart-campus.pdf

Challenge 2 brief description: Software applications for the University of Birmingham, Exchange Building.

In September 2021, the University of Birmingham opened The Exchange. Located in Centenary Square, The Exchange will connect the University's research, teaching and local, national and international networks to create a place of curiosity, celebration, collaboration and change. The Exchange brings together communities to shape the ideas and solutions that will make our cities and regions better places to live, work and learns. Our goal is transformational. We aim to create a place of opportunity to help people, businesses and organisations rethink how cities can be shaped by the people who live and work in them.

The Exchange will achieve our goal by:

- Opening up research through free public exhibitions and community engagement programmes that will enable people from the city and wider region to get involved in the University's work to solve the complex challenges of the 21 century.
- Facilitating data sharing and partnerships that will help grow the regional economy and improve people's quality of life.
- Providing free taster courses, short paid courses and micro-credentials, aimed at widening participation, professional development and/or building regional skills. A particular focus on leadership will encompass the private, public and third sectors and grow the capacity of community leaders.
- The Exchange will also provide a hub to support the development of entrepreneurial skills, helping to create the next generation of businesses.
- Creating an inclusive and unique venue that attracts a range of different audiences and stakeholders to learn with and from each other.

For information on the History and Heritage:

https://www.birmingham.ac.uk/the-exchange/history-and-redevelopment.aspx
http://www.bmbhistory.org.uk/bmb-history_002.htm

We are seeking innovative suggestions for applications of your choice: these could be of direct use or of futuristic nature. *The scope is open and can relate to any application you can envision:*

- **Examples for Challenge 2 may include, but not limited to:**
    - Visitor guide to learn about the Exchange history,
    - VR and Augmented reality applications for virtual events,
    - Events scheduling including seminars, incubation events, public talks
    - App for sponsoring rooms, arts and scientific displays,
    - Software applications for light and colourful patterns projection on the externals of the building,
    - Apps for promoting events and services in the city,
    - System to replicate and live the experience of the old BMB building,
    - Crowdsourcing information system for sharing experiences, personal stories etc,
    - A hologram boot for virtual on-demand event display in the Exchange Building,
    - Virtual/Digital Replica of Exchange in Dubai.

References for challenge 2:
Ref: https://www.birmingham.ac.uk/the-exchange/index.aspx

<u>Challenge 3 Brief Description: Birmingham 2022 Commonwealth Games and the University of Birmingham.</u>

The University of Birmingham is an Official Partner of the Birmingham 2022 Commonwealth Games, having signed the most comprehensive university partnership agreement in the history of the Games. During the Games, which will be held from 28 July to 8 August 2022, the hockey and squash competitions will take place at the University's sporting facilities. The University will also be the main campus village hosting athletes.

We are seeking innovative suggestions for applications of your choice: these could be of direct use. The scope is open and can relate to any application you can envision.

Examples for Challenge 3 may include, but not limited to:

- Crowdsourcing information system for sharing experiences, personal stories about the Commonwealth 2022;
- Virtual international advice forum for Commonwealth 2022 games;
- Ideas for apps and software for COVID and Birmingham 2022 Commonwealth;
- An app that collate and show the information about games, venues, time, etc;
- An app that helps with organising flows of audience in the stadium, to minimise people crowds and reduce chances of covid infection;
- A translation system for people from different countries.

References for Challenge 3:
Ref1: https://www.birmingham2022.com/
Ref2: https://youtu.be/2dVKDKYQDaE

II. Coursework Questions and Instructions – All Challenges:

You software solution has to deal with **ONLY ONE focused application of interest that relate to ONLY ONE of above challenges.** We are seeking innovative suggestions of applications of your choice: these could be of direct use, of futuristic nature, to change behavior etc. The scope is open and can relate to any application you can envision.

The possibilities for Challenge 1, Challenge 2 and Challenge 3 are endless and we are flexible. Your software solution may also aim at educating and engaging with the concerned individuals and/or public. Your application may wish to use AI, Big data and/or emerging technologies (e.g., virtual and augmented reality, gesture controlled interactions, holograms, and/or mobile apps among the others- as your group see it fit) for your chosen system. Please note that the University has been investing in Augmented and Virtual Reality technologies (see that Collaborative engineering lab opposite to the new Engineering School) and you may wish to feature these in your solution.

We are open to *creative, novel ideas and useful applications of societal. We give you the freedom of choice and we do trust your taste!* We hope that your innovative solution will help us showcase the University of Birmingham as a cutting-edge institution in the use of software engineering, mobile apps, data science and/or emerging technologies. We also hope to demonstrate the creativity and ability of our software engineering students to conceptualise innovative digital solutions and to demonstrate their technical skills in systematically engineering data science solutions, covering requirements, distributed architecture and design.

A. Requirements Engineering(Unit 2):

A1. Describe your proposed system in 800 words (or less), stating your own assumptions on the scope of the system. Be precise, concise, and creative! Note - your scope should be reasonable and interesting enough to be handled by group members. **(2 points)**

A2. State the **functional and non-functional requirements** of your system. Use sensible phrasing, grouping and prioritisation as discussed in the sessions. **(2 points)**

B. <u>Software Design with UML(Unit 3):</u>

Use UML to document the analysis and design of your system to a professional level.

B1. Provide a comprehensive Use Case diagram for your system clearly indicating the actors involved. Make sure you use stereotype relations such as <<extend>> and <<include>> in your modelling. **(2 points)**

B2. Choose TWO non-trivial use cases from your Use Case Diagram described in B1. Provide meaningful documentation (i.e. use case specification) for the chosen two Use Cases using pre/post conditions, flow of events. **(2 Points)**

B3. Describe TWO non-trivial scenarios/stories for each of your chosen use cases in B2. **(2 points)**

B4. Provide an Activity Diagram for ONE non-trivial scenario of interest. It doesn't have to be the same one from previous questions. Make sure you use control nodes, swim lanes, and conditions. **(1 point)**

B5. Perform Class Analysis: **(2 point)**

i. Perform noun-verb analysis over your specification:
   Identify the potential classes and potential operations for these classes using what discussed in Lectures.
ii. Derive your CRC cards for the identified classes from the last step.
iii. Produce a First-Cut Class diagram to combine the consolidated results of using the above two techniques – i.e., noun-verb analysis and responsibility-driven analysis.
iv. Detail your Class Diagram. This should provide details on attributes, operations, relationships, visibility, multiplicity, etc) (refer to the class notes).

B6. Provide an Object Diagram for ONE non-trivial scenario of interest. It doesn't have to be the same one from previous questions. **(1 point)**

B7. Provide TWO Sequence Diagrams, each describing a non-trivial scenario of interest. They don't have to be the same ones from previous questions. Make sure you use guard conditions, interaction frames (i.e. envelopes), etc. State any assumption you make. **(2 points)**

B8. Provide TWO State Machine Diagrams, each describing a non-trivial object state transitions or component of the system. Make sure you use

guard conditions, parallel states etc. State any assumption you make. **(2 points)**

For the above questions that require modelling non-trivial scenarios, you may choose from those that were documented as part of the Use Case modelling (B1) and generated scenarios (B3) as this can simplify your work. Please ensure consistency with your description as much as possible.

C. **Software Architecture Style, Modelling and Evaluation (Unit 3 and Unit 4):** You are now a software architect and need to adopt a sensible architecture style for your system. Your group should work on evaluating two possible architecture candidate styles for your solution. Among the numerous possible tradeoffs, you may consider security, performance, scalability, availability, safety, and/or usability etc. once you evaluate your architecture choices and decisions.

C1. Consider two possible candidate architectures styles. Describe each using UML Components Diagrams **(2 points)**

C2. Provide TWO Deployment Diagrams for question C1 **(2 Point)**

Your attempt for C1 and C2 should identify the potential components; specify their interfaces; compose the components; Model ports (if any) and subcomponents (if any).

C3. In no more than 400 words, compare and contrast your two chosen architectures and the tradeoffs. Your answer should provide a sensible conclusion on which of the two is your preferred option. **(2 Points)**

D. Software Testing (Unit 5).

Provide a testing plan of your system, covering SEVEN functional requirements and THREE non-functional. Your plan shall cover the Testing objectives, testing strategy (e.g. black box and white box); and the testing exit criteria. **(2 points)**

E. Usability and Prototyping (Unit 6).

E1. Provide an interactive prototype of your system including FIVE main screens, sketching and conceptualizing the main features of your system. You may use one of the selected tools that will be discussed and demonstrated in the class as part Unit 6; however, be free to use an alternative tools as you see it fit – we are flexible! **(2 points).**

E2. Video recording: Suppose that you have to showcase your prototype and project idea to the mass public and/or to seek funding for a startup opportunity. Provide 3-minutes video-demo of your system to showcase your system to non-experts. Use MP4 format. Be creative! **(2 Points)**

F. <u>Ethics and Professional Practice – (Unit 8) – self-reading:</u>
Your attempt should follow ethical and responsible design. Provide appraisal using the IEEE/ACM software engineering code of ethics. Your appraisal should not exceed 500 words. **(2 points)**

<u>Quality, Consistency and Maturity.</u> Follow a systematic design for all the above questions. Be consistent in using the naming across all your diagrams and their supporting documentations. Strive for a professional analysis and presentation. Make your own judgment on what to be modeled, documented, and presented **(1 point).**

Submission requirements

- The report should be typed and presented in A4 paper format. The covering page should include your group number, ID numbers, and e-mails of all members in your group.
- Your diagrams should be produced using open source UML packages or Visio. Please be free to use any UML package you are comfortable with.
- Please, do work effectively in group towards meeting the requirements of this coursework. **<u>The default meeting mode between group members is virtual to cater for the wide variations of individual circumstances because of COVID. Adhere to UK Governmental guidelines for COVID, in case of guide changes. We will leave it to you to decide on suitable communication platforms for your group (e.g., Skype, Zoom, MS Team etc)</u>**
- Please submit on Dec 6, 11:59 p.m. to Canvas. Go the "Assignment" section in the canvas module, and select the "Group CWK". Only ONE member of your group needed to submit your group report and recorded video. The system will allow you to update your submission before the deadline. The report file type is limited to doc or pdf. The recorded video must be in mp4.
- Almost all the questions of your coursework carry equal weight and computes to 33 points. For your overall mark out of 100, we will multiply your score by 3 and will add a courtesy 1 mark.
- Additional Canvas submission information and moderation would follow separately – <u>please keep watching the space for our announcements.</u>
- We expect to provide virtual feedback support on your coursework (before submission). Please keep watching the Canvas announcements for any additional feedback and organizational support that you will get as part of the coursework.

Enjoyable Coursework and Good Luck!